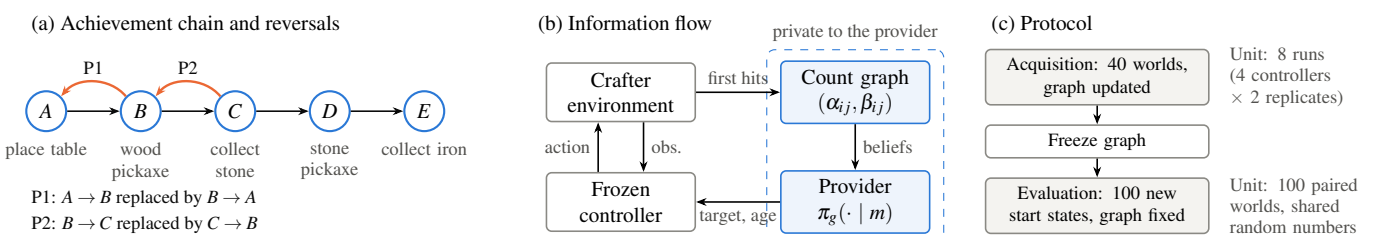
(a) Achievement chain and reversals
P1
P2
A
B
C
D
E
place table
wood pickaxe
collect stone
stone pickaxe
collect iron
P1: $A \to B$ replaced by $B \to A$
P2: $B \to C$ replaced by $C \to B$
(b) Information flow
private to the provider
Crafter environment
first hits
Count graph $(\alpha_{ij}, \beta_{ij})$
action
obs.
beliefs
Frozen controller
target, age
Provider $\pi_g(\cdot \mid m)$
(c) Protocol
Acquisition: 40 worlds, graph updated
Freeze graph
Evaluation: 100 new start states, graph fixed
Unit: 8 runs (4 controllers × 2 replicates)
Unit: 100 paired worlds, shared random numbers